

# When Candidate-Dependent Context Can Leave a Common-Context Robot-Policy Winner Unidentified

Exact regret, graph repair, and finite-sample decisions

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## Abstract

Pairwise robot-policy evaluations may report a global ranking even when each pair is tested in a different physical setting. Even with every pair observed, pair-specific contexts can admit opposite common-context winners. Under primitive pair responses, the constructed full-edge ambiguity set has a unique uniform minimax lottery, with regret  $(K - 1)/(4K)$ . A shared-binary-success sensitivity preserves that lottery and value, but the ambiguity then comes from missing target-context support rather than candidate-dependent routing.

In the shared-success model, identifying all within-context policy differences requires  $m - 1$  comparable bridge types when the route graph has  $m$  components and the allowable bridge graph is connected. Repeating existing routes improves precision, not connectivity.

Simultaneous edge intervals propagate through the same graph constraints to target-difference bounds, possible and certified winners, and a minimax action by linear programming. Under an exact Bradley-Terry target, the winner set is identified exactly when the common-context comparison graph is connected.

Three pinned records show the operational distinction. Complete common-state panels identify exact finite-panel winners in three Ankle tasks. RoboArena’s pair-support graph remains incomplete and lacks a common-context bridge, while TRI’s released outcome tables omit the trial keys needed to reconstruct its published matched-bundle protocol.

## Practitioner summary

- If the evaluated pair can change the task or scene, a precise pairwise leaderboard may still fail to determine which single policy is best for one common deployment population.
- The route graph diagnoses the structural gap: disconnected policy groups need safe, comparable cross-group comparisons in the same declared context.
- In the shared-success model,  $m - 1$  bridge types identify all within-context differences across  $m$  components when allowable bridges connect them.
- With finite samples, use jointly valid edge intervals; the graph program inherits their coverage but cannot repair invalid sampling units.
- Connectivity does not ensure precision, validity, or safety.

## 1. The decision precedes the score

A robot-policy comparison can support several distinct actions:

1. choose one policy for a common deployment population;
2. choose between two policies presented under the same pair-specific mechanism;

3. make a same-mechanism choice within each future presented pair; or
4. report a mechanism-specific tournament score.

These actions are not interchangeable. A complete table of pairwise win probabilities identifies a tournament under the mechanisms that generated its edges. It need not identify how the policies compare in a common physical context population. The distinction matters when the evaluated pair can change the task, scene, operator behavior, or other context.

We distinguish identification, minimax decisions under ambiguity, and evidence connecting the target to a real system. We apply established ideas from partial identification, minimax regret, randomized decisions, Borda ranking, comparison graphs, and target-first evaluation to candidate-dependent physical contexts.

The closest antecedents already establish interval decisions, graph-based score identifiability and precision, and robust lotteries. Our scoped contribution is their interaction here: an exact candidate-dependent-context construction, its edge-coupled regret values, a route-colored decision program, and an executable finite-sample interface.

## 2. Pair-first context can leave the global action unidentified

### 2.1 Target and observed routes

Let  $K \geq 3$  policies be indexed by  $i = 1, \dots, K$ , and let one shared context variable  $C$  have the target distribution. For a common-context target, write  $q_{ij}$  for policy  $i$ 's expected half-credit comparison score against  $j$ , with  $q_{ji} = 1 - q_{ij}$  and  $q_{ii} = 1/2$ . This score may be a conditional comparison probability or a deterministic fractional score. The unrestricted construction below treats pair responses as primitive and imposes no shared scalar-outcome, random-utility, stochastic-transitivity, or cross-edge coherence constraint. Under the uniform-reference normalized Borda target,

$$V_i = \frac{\frac{1}{2} + \sum_{j \neq i} q_{ij}}{K}.$$

Borda is the declared comparison target. It equals success-mean ordering in the shared-binary model below, but is not a generic deployment utility.

Now suppose the policy pair is selected before context is constructed. The observed edge for pair  $\{i, j\}$  averages outcomes under a pair-specific context mechanism  $G_{ij}$ , not necessarily under the common target population.

### 2.2 Observational-equivalence result

Let the one shared context variable take values A and B with equal target weight. Each pair has a route  $G_{ij}$ , degenerate at one of those shared contexts. Fix the complete routed score law as a deterministic half tie. Choose the hidden context-specific primitive pair score so that the target edge can take any value in  $[1/4, 3/4]$ . These context weights differ from Theorem 3's opponent-reference weights. All completions generate the same observed route/context/score law. Two explicit completions reverse the unique common-context winner.

For every finite  $K \geq 3$ , one complete-support pair-first half-win law is compatible with the full primitive-response box

$$q_{ij} = \frac{1}{2} + \delta_{ij}, \quad \delta_{ij} \in [-1/4, 1/4],$$

independently for all  $\binom{K}{2}$  unordered pairs.

**Theorem 1 (construction-bounded non-identification).** For every  $K \geq 3$ , there is a pair-first, complete-pair-support observed law for which all compatible worlds agree on every observed route but the

common-context Borda winner is not identified. Repeating the same routed comparisons without a bridge does not shrink this compatible target box.

Construction and proof: [Supplement S1](#). This counterexample does not apply to every pair-first design.

### 3. Exact minimax-regret solutions for the constructed edge box

#### 3.1 Uniform opponent reference

Let  $p$  be a policy lottery, used here as an ambiguity benchmark rather than a deployment recommendation, chosen before the compatible world and policy draw. Its worst-compatible expected regret is

$$\mathcal{R}(p) = \sup_q \left\{ \max_w V_w - \sum_i p_i V_i \right\}.$$

For a fixed candidate winner  $w$ , independent edge extrema yield

$$\mathcal{R}(p) = \frac{1}{4K} \max_w \sum_{i < j} |\{1(i = w) - p_i\} - \{1(j = w) - p_j\}|.$$

**Theorem 2 (full-edge uniform-reference minimax).** In this compatible box:

- every deterministic singleton has worst-compatible regret  $(K - 1)/(2K)$ ;
- the unique minimax lottery is uniform over all  $K$  policies;
- its worst-compatible expected regret is  $(K - 1)/(4K)$ .

Proof: [Supplement S2](#).

The factor of two concerns ex-ante expected regret. It does not halve the worst regret of the realized policy.

Each policy's marginal interval has width  $(K - 1)/(2K)$ , which already gives the deterministic regret in Stoye's arbitrary-treatment interval model. An independent product box would give uniform-lottery regret  $(K - 1)^2/(2K^2)$ . Our smaller value comes from antisymmetric edge coupling: states with one policy at its upper bound and every other policy at its lower bound are infeasible. Supplement S6 gives the comparison.

#### 3.2 Arbitrary nonnegative opponent reference

Let nonnegative reference weights  $r_i$  sum to one and define

$$V_i = \sum_j r_j q_{ij}.$$

Order the three smallest weights as  $a = r_{(1)} \leq b = r_{(2)} \leq g = r_{(3)}$ . The robust objective reduces to

$$\mathcal{R}(p) = \frac{1}{4} \max_w \left[ 1 - p_w + \sum_{\substack{i < j \\ i, j \neq w}} |r_j p_i - r_i p_j| \right].$$

**Theorem 3 (weighted-reference minimax).** The exact minimax value is

$$\mathcal{R}^* = \frac{2 - a - b}{8}.$$

The minimizers form the line segment in Supplement S3 and are unique exactly when the second- and third-smallest reference weights are equal.

**Corollary 3.1 (hard support exclusion).** If the lottery must assign zero mass to zero-reference policies while those policies remain possible oracle comparators,  $p = r$  is uniquely minimax with value  $1/4$  whenever any reference weight is zero (Supplement S3).

Here  $r$  is the opponent-reference distribution, not the pair-sampling distribution or policy lottery  $p$ . Its empirical choice lies outside the theorem.

### 3.3 Shared-success sensitivity

Under a shared binary success outcome  $Y_i(C)$ , a pair receives one point for a strict success win and one half for a tie. Route every observed pair through shared context A with identical policy outcomes, and let  $x_i = E[Y_i(B)] \in [0, 1]$ . Then

$$q_{ij} = \frac{1}{2} + \frac{1}{4}(x_i - x_j).$$

The compatible edges form a gradient polytope satisfying  $\delta_{ik} = \delta_{ij} + \delta_{jk}$ , not the full independent box. Nevertheless,  $x = e_1$  and  $x = e_K$  give opposite unique common-context winners while preserving the complete observed half-tie law. For any opponent-reference distribution  $r$ ,

$$V_i = \frac{1}{2} + \frac{1}{4}(x_i - r^\top x), \quad \mathcal{R}(p) = \frac{1}{4}(1 - \min_i p_i).$$

Thus deterministic worst regret becomes  $1/4$ , while the unique uniform minimax lottery and value  $(K - 1)/(4K)$  are unchanged and  $r$  cancels. Theorem 3's weighted optimizer face and Corollary 3.1's uniqueness are therefore response-model dependent. This reflects missing target-context support, not candidate-dependent routing (Supplement S4).

If the observed-context success profile has range  $D < 1$ , any policy can be uniquely best under a compatible missing-context completion within the shared-success model (Supplement S4).

### 3.4 Candidate-dependent route graphs

For each positive-target-weight context  $c$ , form a route graph whose edges are the pairs observed there. Policy-success means are identified up to one additive offset per component. Connected graphs identify all contextwise differences; disconnected components can retain target-relevant offsets when the outcome bounds leave slack. The exact criterion is zero compatible width for every target difference, not connectivity alone (Supplement S4).

In a three-policy example, pair 1–2 routes through B while 1–3 and 2–3 route through A. The routed law is compatible with unique common-target winners 1 and 3. Its ex-ante minimax lottery is uniquely  $(2/3, 0, 1/3)$ , with regret  $1/12$ , rather than the uniform lottery above. The corresponding minimax action is a linear program (Supplement S4).

If a context graph has  $m$  components,  $m - 1$  cross-component pair types are necessary and sufficient when the allowable-pair quotient graph is connected; with costs, a minimum spanning tree gives the cheapest repair (Supplement S4).

#### 3.4.1 Hypothetical robot design

Suppose policies  $A, B, C$  are evaluated in near and far reset strata. Near comparisons  $A - C$  and  $B - C$  connect the roster, but far contains only  $A - B$ . Adding either safe  $C - A$  or  $C - B$  far-stratum comparison closes the gap. The bridge requires comparable randomized trials, recorded policy and reset lineage, and a separate precision analysis.

### 3.4.2 Finite samples

Replace each exact edge equation by a simultaneous interval,  $\ell_c \leq E_c x^c \leq u_c$ . Linear programs then give sharp bounds on every target difference, the policies that can still win, any winner certified against all compatible values, and the minimax lottery. If the edge intervals jointly cover with probability  $1 - \alpha$ , these projected statements inherit that coverage. They do not repair dependence, adaptive sampling, or target drift (Supplement S4.3).

### 3.4.3 Bradley–Terry structure

Suppose exact common-context comparisons obey  $q_{ij} = \sigma(s_i - s_j)$ . Applying the inverse link to each observed edge identifies score differences along paths. The Borda-winner set is therefore identified exactly when the comparison graph is connected; a unique winner also requires a unique largest score. Disconnected components retain separate offsets and need bridges. This is an established graph-identifiability consequence, not a new Bradley–Terry result (Supplement S4.4).

## 3.5 Relation to maximal and robust lotteries

On the observed zero margin matrix, every lottery is maximal. Robust maximal and Borda-regret rules both select the uniform lottery here but use different losses and adversaries (Supplement S5).

## 4. A different action can remain identified

For each pair  $a < b$ , define the mechanism-specific edge

$$\theta_{ab}^{pc} = E_{C \sim G_{ab}}[Z_{ab}(C)].$$

If a future unordered pair is drawn from a fixed outcome-independent law  $\pi$ , and the same  $G_{ab}$  will be used, a routing rule has value

$$V^{pc}(\delta) = \sum_{a < b} \pi_{ab} [\mathbf{1}\{\delta(a, b) = a\} \theta_{ab}^{pc} + \mathbf{1}\{\delta(a, b) = b\} (1 - \theta_{ab}^{pc})].$$

The objective is separable, so the optimal same-mechanism within-pair choice selects the locally preferred member of each positive-weight pair.

In a three-edge example, the local rule has value 3/4, while always choosing the lower-index policy has value 7/12, even though all tournament scores tie at 1/2. This target requires a fixed future pair law, positive edge support, stable pair mechanisms, and assignment-aware uncertainty (Supplement S5).

## 5. What public records can identify

We fixed three public releases before aggregate analysis. They are purposive examples, and missing public fields do not imply private absence.

### 5.1 A positive finite-panel case

Three AnkIle R5 releases each run three fixed policy artifacts on the same 50 declared Sobol states: routing, marker, and square ([routing](#); [marker](#); [square](#)). All 450 policy-state cells are present. The success counts are ((8,9,12)), ((19,32,31)), and ((28,36,33)), so the exact released-panel winner is identified in each task, by margins of 3, 1, and 3 successes. Each three-edge route graph is complete.

This identifies only the displayed states. One rollout per cell supplies no population uncertainty law, and the files do not reconstruct achieved reset acceptance or the full retry ledger. The omission can matter at

the scale of the reported differences. Under a worst-case sensitivity that replaces an entire retained three-policy round by another binary outcome vector, reversing the released winner requires only two routing rounds, one marker round, or two square rounds. This is not evidence that any such replacement occurred. The released configuration enables incomplete-round reruns for routing and marker, but not square, so the retained attempt history is needed to distinguish a stable result from selection through reruns.

## 5.2 Pair support is not a common-context graph

The pinned RoboArena release contains 3,883 sessions, 21 policy labels, and 104 of 210 possible label pairs ([data](#); [paper](#)). One unpaired label forms a separate component; the other 20 labels are connected. Twenty-seven observed edges appear in only one session.

This is policy-pair support, not the contextwise route graph of Section 3.4. Tasks and scenes are constructed for the selected pair, and the release does not bind assignment weights or pool epochs, exact initial states and resets, robot instances, or retry ancestry to a common target. Connectivity among the 20 compared labels therefore does not identify a common-context winner.

## 5.3 A strong protocol can lose its join at release

TRI reports 1,800 real-world rollouts in blind, randomized test bundles that hold the initial condition across compared policies ([paper](#); [data](#)). Dryad version 4 releases outcome arrays, but no bundle, rollout, initial-condition, realized-order, reset, session, robot, retry, or immutable policy-version identifier. Two published success rates also disagree with their own arrays and counts. The public tables support marginal checks, not reconstruction of the matched route edges.

Supplement S7 compares five additional public systems using the same evidence layers.

# 6. What the missing records must answer

For a finite common-state policy comparison, the practical questions are simple: were the declared states actually achieved comparably for every policy, which attempts were retained or replaced, and could that choice depend on policy or observed performance? An attempt record needs only enough fields to join state, policy and realized order to reset status, execution status, outcome, and any replacement. Video, operator identity, and fine-grained telemetry are not required for this check.

For AnkIle, the decisive reanalysis is to compare the published retained panel with the first valid attempt per policy-state cell and with all valid attempts, averaged within each cell before restoring the original equal weighting of the 50 states. This prevents frequently retried cells from receiving extra weight. If the winner is stable, the provenance strengthens the finite-panel claim. If it changes, the size and direction of the retention effect should be reported. For a disconnected route graph, the substantive repair remains a safe common-context bridge; repeating an existing route cannot identify the missing comparison.

## 7. Limitations

These counterexamples do not characterize every pair-first design or recommend a policy lottery. Theorems 2-3 use a primitive-response box; the shared-success sensitivity uses a narrower model. The real records are purposive. AnkIle identifies only three exact released panels; its thin winner margins are sensitive to one or two worst-case matched-round replacements, and the public files do not show whether such replacements occurred. RoboArena supplies pair support but not a common-context target, and TRI's public tables do not retain its matched-bundle join.

The finite-sample program inherits the validity of supplied simultaneous edge intervals. It does not provide a sampling unit, power, minimum detectable effect, or stopping rule; each bridge study still needs those design choices.

The English-language review does not claim priority for the underlying methods.

## Data and code availability

Data, code, and proofs are available in the package and supplement. Project materials: <https://github.com/trilamsr/research/tree/main/sim2real-correlation-audit>.

## Author and declarations

Tri Lam developed the analysis, conducted the public-record review, and wrote the manuscript. No conflicts of interest are declared. The paper reports no new human- or animal-subject experiment.

## References

- Aziz, H. et al. (2015). *Possible and Necessary Winners of Partial Tournaments*. JAIR 54. <https://doi.org/10.1613/jair.4856>
- Atreya, P. et al. (2025). *RoboArena: Distributed Real-World Evaluation of Generalist Robot Policies*. PMLR 305. <https://proceedings.mlr.press/v305/atreya25a.html>
- Anwar, A. et al. (2025). *Efficient Evaluation of Multi-Task Robot Policies With Active Experiment Selection*. <https://arxiv.org/abs/2502.09829>
- Balduzzi, D. et al. (2018). *Re-evaluating Evaluation*. NeurIPS 31. <https://arxiv.org/abs/1806.02643>
- Binette, O. and Reiter, J. P. (2024). *Improving the Validity and Practical Usefulness of AI/ML Evaluations Using an Estimands Framework*. <https://arxiv.org/abs/2406.10366>
- Bauer, S. et al. (2022). *A Robot Cluster for Running Reinforcement Learning Experiments on Real Robots*. CoRL. <https://proceedings.mlr.press/v176/bauer22a.html>
- Brandl, F., Brandt, F., and Seedig, H. G. (2016). *Consistent Probabilistic Social Choice*. *Econometrica* 84(5). <https://doi.org/10.3982/ECTA13337>
- Brill, M., Freeman, R., and Conitzer, V. (2016). *Computing Possible and Necessary Equilibrium Actions (and Bipartisan Set Winners)*. AAAI. <https://doi.org/10.1609/aaai.v30i1.10052>
- Dudík, M. et al. (2015). *Contextual Dueling Bandits*. PMLR 40. <https://proceedings.mlr.press/v40/dudik15.html>
- Hajek, B., Oh, S., and Xu, J. (2014). *Minimax-Optimal Inference from Partial Rankings*. NeurIPS 27. <https://arxiv.org/abs/1406.5638>
- Jiang, T. et al. (2026). *What Are We Actually Benchmarking in Robot Manipulation?* <https://arxiv.org/abs/2606.04233>
- Kress-Gazit, H. et al. (2024). *Robot Learning as an Empirical Science: Best Practices for Policy Evaluation*. <https://arxiv.org/abs/2409.09491>
- Kress-Gazit, H. et al. (2026). *A Careful Examination of Large Behavior Models for Multitask Dexterous Manipulation*. <https://arxiv.org/abs/2507.05331>
- Khalaf, H. et al. (2026). *Robust AI Evaluation through Maximal Lotteries*. <https://arxiv.org/abs/2602.21297>
- Lanctot, M. et al. (2025 version). *Evaluating Agents using Social Choice Theory*. <https://arxiv.org/abs/2312.03121>
- Lanctot, M. et al. (2026). *Active Evaluation of General Agents*. <https://arxiv.org/abs/2601.07651>
- Lu, T. and Boutilier, C. (2011). *Robust Approximation and Incremental Elicitation in Voting Protocols*. IJCAI. [https://www.cs.toronto.edu/~cebly/Papers/LuBoutilier\\_Elicitation\\_ijcai11.pdf](https://www.cs.toronto.edu/~cebly/Papers/LuBoutilier_Elicitation_ijcai11.pdf)
- Montiel Olea, J. L., Qiu, C., and Stoye, J. (2025 version). *Decision Theory for Treatment Choice Problems with Partial Identification*. <https://arxiv.org/abs/2312.17623>
- Osting, B., Brune, C., and Osher, S. (2014). *Optimal Data Collection for Informative Rankings Expose Well-Connected Graphs*. JMLR 15(85). <https://www.jmlr.org/papers/v15/osting14a.html>
- Manski, C. F. (2009). *Diversified Treatment under Ambiguity*. <https://doi.org/10.1111/j.1468-2354.2009.00558.x>
- PhAIL (2026). *PhAIL v1.0*. arXiv:2605.29710v1. <https://arxiv.org/abs/2605.29710>
- RoboDojo (2026). *RoboDojo*. arXiv:2607.04434v3. <https://arxiv.org/abs/2607.04434>

- Saha, A., Koren, T., and Mansour, Y. (2021). *Adversarial Dueling Bandits*. PMLR 139. <https://proceedings.mlr.press/v139/saha21a.html>
- Selvaraj, P., Uttini, L., and Kuosmanen, V. (2026). *ArmnetBench v0.1: Parallel Real-World Evaluation of Manipulation Policies on a Low-Cost Arm Farm*. <https://arxiv.org/abs/2607.24481>
- Shah, N. and Wainwright, M. (2018). *Simple, Robust and Optimal Ranking from Pairwise Comparisons*. JMLR 18(199). <https://www.jmlr.org/papers/v18/16-206.html>
- Shah, N., Balakrishnan, S., Bradley, J., Parekh, A., Ramchandran, K., and Wainwright, M. (2016). *Estimation from Pairwise Comparisons: Sharp Minimax Bounds with Topology Dependence*. JMLR 17(58). <https://www.jmlr.org/papers/v17/15-189.html>
- Skalse, J. et al. (2023). *Invariance in Policy Optimisation and Partial Identifiability in Reward Learning*. PMLR 202. <https://proceedings.mlr.press/v202/skalse23a.html>
- Stoye, J. (2009). *Partial Identification and Robust Treatment Choice: An Application to Young Offenders*. <https://doi.org/10.1080/15598608.2009.10411923>
- Stoye, J. (2007). *Minimax Regret Treatment Choice with Incomplete Data and Many Treatments*. *Econometric Theory* 23(1). <https://doi.org/10.1017/S0266466607070089>
- Stoye, J. (2012). *Minimax Regret Treatment Choice with Covariates or with Limited Validity of Experiments*. *Journal of Econometrics* 166(1). <https://doi.org/10.1016/j.jeconom.2011.06.012>
- Suk, J. and Agarwal, A. (2024). *Non-Stationary Dueling Bandits Under a Weighted Borda Criterion*. <https://arxiv.org/abs/2403.12950>
- UMI-Bench (2026). *UMI-Bench 1.0*. arXiv:2606.10382v1. <https://arxiv.org/abs/2606.10382>
- Viappiani, P. (2020). *Robust Winner Determination in Positional Scoring Rules with Uncertain Weights*. *Theory and Decision* 88. <https://doi.org/10.1007/s11238-019-09734-3>

# Supplement to: When Candidate-Dependent Context Can Leave a Common-Context Robot-Policy Winner Unidentified

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## Status and notation

This supplement gives the derivations for the three construction-bounded theorems, the route-graph uncertainty program, and the Bradley–Terry connectivity corollary in the main paper.

There are  $K \geq 3$  policies. For  $i < j$ ,  $q_{ij}$  is policy  $i$ 's expected half-credit comparison score against  $j$  in the target context population;  $q_{ji} = 1 - q_{ij}$  and  $q_{ii} = 1/2$ . Under uniform opponent reference,

$$V_i = \frac{\frac{1}{2} + \sum_{j \neq i} q_{ij}}{K}.$$

For the compatible class below,

$$q_{ij} = \frac{1}{2} + \delta_{ij}, \quad \delta_{ij} \in [-1/4, 1/4], \quad \delta_{ji} = -\delta_{ij}.$$

## S1. Compatible worlds and Theorem 1

### S1.1 Explicit potential-outcome schedule

Let one shared context variable  $C$  take values A and B, and define the common target population as their equal-weight mixture. For every unordered pair  $e = \{i, j\}$ , choose one value as its routed context  $R_e$ . The route may alternate across edges; only its being fixed independently of the compatible target completion matters. Pair scores are primitive responses: this construction imposes no shared scalar policy outcome or other cross-edge coherence restriction.

For any desired target edge  $q_e \in [1/4, 3/4]$ , define the potential outcomes for the orientation  $i < j$  by

$$E[Z_e \mid R_e] = \frac{1}{2}, \quad E[Z_e \mid \bar{R}_e] = 2q_e - \frac{1}{2}.$$

The second outcome is valid because

$$0 \leq 2q_e - \frac{1}{2} \leq 1.$$

Fix the complete routed score law as a deterministic half tie, not merely an equality of routed means. The observed routed edge is therefore  $1/2$ , while the equal-context target projection is

$$\frac{1}{2} \left\{ \frac{1}{2} + (2q_e - \frac{1}{2}) \right\} = q_e.$$

This construction is applied separately to every one of the  $\binom{K}{2}$  edges. Consequently every vector  $(q_{ij} : i < j) \in [1/4, 3/4]^{\binom{K}{2}}$  has a valid potential-outcome schedule, and every such schedule projects to the same observed complete-pair-support half-win law.

## S1.2 Opposite unique winners

Two endpoint completions are enough to show that the common-context winner is not identified. If  $q_{ij} = 3/4$  for every  $i < j$ , policy 1 has strictly the largest Borda value. If  $q_{ij} = 1/4$  for every  $i < j$ , policy  $K$  has strictly the largest value. More explicitly, under the all-high completion the unnormalized score of policy  $i$  is

$$\frac{1}{2} + \frac{3}{4}(K - i) + \frac{1}{4}(i - 1),$$

which strictly decreases with  $i$ . Under the all-low completion the coefficients  $1/4$  and  $3/4$  exchange, so the score strictly increases with  $i$ . The two worlds agree on every observed pair, routed context, and routed outcome.

Thus, for every  $K \geq 3$ , this particular complete-support observed law is compatible with opposite unique common-context winners. Repeating routed comparisons estimates the routed half-win law more precisely but supplies no information about the unobserved context-specific potential outcomes.

## S2. Uniform-reference edge-coupled minimax

Let  $p$  be an ex-ante policy lottery, and for a candidate oracle winner  $w$  put

$$c_i = \mathbf{1}\{i = w\} - p_i.$$

Because  $\sum_i c_i = 0$ , the midpoint terms cancel. The regret against candidate  $w$  is

$$V_w - \sum_i p_i V_i = \frac{1}{K} \sum_{i < j} (c_i - c_j) \delta_{ij}.$$

The edges vary independently over  $[-1/4, 1/4]$ , so maximizing each term at its sign-compatible endpoint gives

$$\mathcal{R}(p) = \frac{1}{4K} \max_w \sum_{i < j} |c_i - c_j|. \quad (\text{S2.1})$$

### S2.1 Deterministic and uniform values

For a deterministic choice  $p = e_s$ , choosing  $w \neq s$  in (S2.1) gives one difference of magnitude two,  $2(K - 2)$  differences of magnitude one, and all other differences zero. Hence

$$\mathcal{R}(e_s) = \frac{2 + 2(K - 2)}{4K} = \frac{K - 1}{2K}.$$

For the uniform lottery,  $p_i = 1/K$ . For any  $w$ , only the  $K - 1$  edges incident to  $w$  have nonzero differences, all of magnitude one. Therefore

$$\mathcal{R}(p_{\text{unif}}) = \frac{K - 1}{4K}.$$

These are ex-ante expected-regret values. The second expression does not bound the worst regret of the policy realized from the lottery.

## S2.2 Uniqueness

For a general  $p$ , averaging the unscaled objective in (S2.1) over all candidate winners gives

$$\frac{1}{K} \sum_w \sum_{i < j} |c_i - c_j| = K - 1 + \frac{K - 2}{K} \sum_{i < j} |p_i - p_j|. \quad (\text{S2.2})$$

The maximum over  $w$  is at least this average. Equation (S2.2) equals  $K - 1$  only when every  $p_i$  is equal, and is strictly larger otherwise. The uniform lottery attains  $K - 1$  for every  $w$ , so it is the unique minimizer.

## S3. Weighted-reference minimax

Now let  $r_i \geq 0$ ,  $\sum_i r_i = 1$ , and define

$$V_i = \sum_j r_j q_{ij}.$$

For candidate winner  $w$ , use the same  $c_i = \mathbf{1}\{i = w\} - p_i$ . Maximizing each edge over the compatible interval gives

$$\mathcal{R}(p) = \frac{1}{4} \max_w F_w,$$

where the incident-edge terms sum to  $1 - p_w$ , leaving

$$F_w = 1 - p_w + \sum_{\substack{i < j \\ i, j \neq w}} |r_j p_i - r_i p_j|. \quad (\text{S3.1})$$

Order the weights and relabel so that

$$a = r_1 \leq b = r_2 \leq g = r_3 \leq \dots.$$

Tie permutations do not change the resulting optimizer set.

### S3.1 Lower bound

The first two candidate-winner objectives obey

$$F_1 + F_2 \geq 2 - p_1 - p_2 + |p_1 + p_2 - a - b| \geq 2 - a - b. \quad (\text{S3.2})$$

At least one of  $F_1, F_2$  is therefore at least  $(2 - a - b)/2$ , which yields

$$\mathcal{R}^* \geq \frac{2 - a - b}{8}. \quad (\text{S3.3})$$

### S3.2 Equality conditions and complete optimizer face

To expose the equality cases, write  $C = 1 - a - b$  and  $s = p_1 + p_2$ . Before the final inequality in (S3.2), the complete lower-bound chain is

$$\begin{aligned}
F_1 + F_2 &\geq 2 - s + \sum_{k \geq 3} \{|r_k p_2 - b p_k| + |r_k p_1 - a p_k|\} \\
&\geq 2 - s + \sum_{k \geq 3} |r_k s - (a + b) p_k| \\
&\geq 2 - s + |C s - (a + b)(1 - s)| \\
&= 2 - s + |s - a - b| \\
&\geq 2 - a - b.
\end{aligned} \tag{S3.2a}$$

The first line drops twice the nonnegative terms  $|r_j p_i - r_i p_j|$  for  $3 \leq i < j$ . If  $p$  is optimal, both  $F_1$  and  $F_2$  are at most  $(2 - a - b)/2$ , so equality must hold throughout (S3.2a). The dropped terms must therefore vanish. Because the tail has positive total reference weight, this implies, for one common  $\lambda$ ,

$$p_i = \lambda r_i \quad \text{for every } i \geq 3. \tag{S3.2b}$$

In particular, any zero-weight tail index has  $p_i = 0$ ; the argument never divides by a zero reference weight. Equality in the last line requires  $s \geq a + b$ . Since  $\lambda = (1 - s)/C$ , equality in the two triangle inequalities further requires

$$p_1 - a\lambda \geq 0, \quad p_2 - b\lambda \geq 0. \tag{S3.2c}$$

The two candidate-winner objectives must also be equal:

$$F_1 = F_2 = \frac{2 - a - b}{2}.$$

Put

$$h = p_1 - \frac{a + b}{2}.$$

Solving  $F_1 = (2 - a - b)/2$ , (S3.2b), and the simplex constraint gives

$$\begin{aligned}
p_1 &= \frac{a + b}{2} + h, \\
p_2 &= \frac{b(2 - a - b)}{2(1 - a)} + \frac{1 - b}{1 - a} h,
\end{aligned}$$

and, for  $i \notin \{1, 2\}$ ,

$$p_i = r_i \frac{2 - a - b}{1 - a} \left( \frac{1}{2} - \frac{h}{1 - a - b} \right). \tag{S3.4}$$

In this parametrization,

$$p_2 - b\lambda = \frac{h}{C},$$

so (S3.2c) forces  $h \geq 0$ ; the other sign condition is then satisfied because  $b \geq a$ . Only on this equality-case domain, substitution in (S3.1) gives, for every  $j \geq 3$ ,

$$F_j - \frac{2-a-b}{2} = 2h - (r_j - b). \quad (\text{S3.5})$$

Thus all remaining candidate-winner inequalities are satisfied exactly when

$$0 \leq h \leq \frac{g-b}{2}.$$

Every point in this segment attains the lower bound (S3.3), so

$$\mathcal{R}^* = \frac{2-a-b}{8}.$$

The segment collapses to a single point exactly when  $b = g$ . The proportional water-filling solution is the  $h = 0$  endpoint and has no general priority over the rest of the segment.

When exactly two weights are zero, they are indices 1 and 2 and receive equal mass  $h$ , up to half the smallest positive reference weight. When at least three weights are zero,  $b = g = 0$ ; the segment collapses to  $h = 0$ , and  $p = r$  is unique.

### S3.3 Hard support exclusion

Theorem 3 allows  $p_i > 0$  when  $r_i = 0$ . Now restrict the action lottery by  $r_i = 0 \Rightarrow p_i = 0$ , while retaining zero-reference policies as possible oracle comparators. For any zero-reference candidate winner  $z$ , (S3.1) gives

$$F_z = 1 + \sum_{\substack{i < j \\ r_i > 0, r_j > 0}} |r_j p_i - r_i p_j| \geq 1. \quad (\text{S3.6})$$

The feasible lottery  $p = r$  makes the sum zero and every other  $F_w \leq 1$ . Equality forces  $p_i = \lambda r_i$  on positive support, and normalization gives  $\lambda = 1$  (support size one is immediate). Hence the exact value is  $1/4$ , uniquely at  $p = r$ . Relative to Theorem 3, exactly one zero raises the value by  $r_{(2)}/8$ ; exactly two retain only  $h = 0$ ; at least three change nothing.

## S4. Shared-success sensitivity and route-graph repair

Let  $Y_i(C) \in \{0, 1\}$  be policy  $i$ 's success outcome in shared context  $C$ . Give policy  $i$  one point for a strict success win over policy  $j$ , zero for a strict loss, and one half for a tie:

$$Z_{ij}(C) = \mathbf{1}\{Y_i(C) > Y_j(C)\} + \frac{1}{2}\mathbf{1}\{Y_i(C) = Y_j(C)\}.$$

For any joint binary law,

$$\begin{aligned} E[Z_{ij}(C)] &= P(Y_i = 1, Y_j = 0) + \frac{1}{2}P(Y_i = Y_j) \\ &= \frac{1}{2} + \frac{1}{2}\{E[Y_i(C)] - E[Y_j(C)]\}. \end{aligned} \quad (\text{S4.1})$$

No independence between  $Y_i$  and  $Y_j$  is required for (S4.1). Route every observed pair through shared context A and fix identical policy outcomes there, so the complete observed score law is a deterministic half tie. Because every pair uses the same route, this section isolates absence of support for target context B; it does not test candidate-dependent routing. Put  $x_i = E[Y_i(B)] \in [0, 1]$ . Under the equal-context target,

$$q_{ij} = \frac{1}{2} + \frac{1}{4}(x_i - x_j). \quad (\text{S4.2})$$

Thus  $\delta_{ij} = (x_i - x_j)/4$ , and every triple satisfies

$$\delta_{ik} = \delta_{ij} + \delta_{jk}. \quad (\text{S4.3})$$

The compatible region is a gradient image of  $[0, 1]^K$ , not the full  $\binom{K}{2}$ -dimensional edge box. Yet  $x = e_1$  and  $x = e_K$  preserve the observed law and make policies 1 and  $K$ , respectively, unique target Borda winners for every  $K \geq 3$ .

For arbitrary nonnegative opponent-reference weights  $r$  summing to one,

$$\begin{aligned} V_i &= \sum_j r_j \left\{ \frac{1}{2} + \frac{1}{4}(x_i - x_j) \right\} \\ &= \frac{1}{2} + \frac{1}{4}(x_i - r^\top x). \end{aligned} \quad (\text{S4.4})$$

The reference term is common to all actions and cancels from regret:

$$\begin{aligned} \mathcal{R}(p) &= \frac{1}{4} \sup_{x \in [0, 1]^K} \{ \max_i x_i - p^\top x \} \\ &= \frac{1}{4} \max_w (1 - p_w) = \frac{1}{4} (1 - \min_i p_i). \end{aligned} \quad (\text{S4.5})$$

The last equality is sharp at a cube vertex with  $x_w = 1$  and every other coordinate zero. Maximizing the smallest lottery mass uniquely gives  $p_i = 1/K$ , hence

$$\mathcal{R}^* = \frac{K-1}{4K}. \quad (\text{S4.6})$$

Every deterministic lottery has a zero-mass alternative and regret  $1/4$ . The structured and primitive models therefore share the unique uniform lottery and its value, but not deterministic regret, weighted optimizer geometry, or the hard-support uniqueness statement.

The half-tie profile is one point in a larger exact class. Let  $a_i = E[Y_i(A)]$  be an arbitrary compatible observed-context profile, defined up to a common offset by the routed pair scores, and retain  $x_i = E[Y_i(B)]$ . Then

$$o_{ij} = \frac{1}{2} + \frac{1}{2}(a_i - a_j), \quad q_{ij} = \frac{1}{2} + \frac{1}{4}\{(a_i + x_i) - (a_j + x_j)\}. \quad (\text{S4.7})$$

For one pair, the missing-context score ranges over  $[0, 1]$ , so

$$q_{ij} \in [o_{ij}/2, (o_{ij} + 1)/2]. \quad (\text{S4.8})$$

The interval always has width  $1/2$  and strictly straddles one half exactly when  $o_{ij} \in (0, 1)$ . Globally, put  $D = \max_i a_i - \min_i a_i$ . If  $D < 1$ , the completion  $x = e_w$  makes policy  $w$  uniquely best for every  $w$ , since  $a_w + 1 > a_j$  for all  $j \neq w$ . If  $D = 1$ , any policy attaining  $\min a$  can at best tie a maximum- $a$  policy and cannot be a unique winner. Thus strict interiority is an exact boundary.

For heterogeneous  $a$ , fixed-winner maximization over the missing-context cube gives

$$\begin{aligned}
\mathcal{R}(p; a) &= \frac{1}{4} \max_w \sup_{x \in [0,1]^K} \{a_w + x_w - p^\top(a + x)\} \\
&= \frac{1}{4} \{1 - p^\top a + \max_w (a_w - p_w)\}.
\end{aligned} \tag{S4.9}$$

A common offset in  $a$  cancels. Constant  $a$  reduces (S4.9) to (S4.5). The result is relative-open only within the additive shared-success law with equal target-context weights; it is neither ambient-open in unrestricted pair-score space nor a finite-sample confidence statement.

### S4.1 Route-colored graph extension

For context  $c$ , orient the observed route edges and let  $E_c$  be the incidence matrix. The routed half-credit scores identify  $E_c x^c = d_c$ ,  $0 \leq x^c \leq 1$ . The null space has one constant-offset degree of freedom per connected component. Consequently, if every positive-target-weight route graph is connected, every within-context policy difference and hence every common-target difference is identified. Disconnection can retain component offsets when the box constraints leave slack; the exact criterion is zero compatible width for every target difference  $\mu_i - \mu_j$ , where  $\mu = \sum_c \alpha_c x^c$ .

For a lottery  $p$ , fixed opponent-reference terms cancel and

$$\mathcal{R}(p) = \frac{1}{2} \max_w \sup_{\{x^c\}} (e_w - p)^\top \sum_c \alpha_c x^c. \tag{S4.10}$$

Each inner support function is linear. Introducing free  $\lambda_{wc}$  and  $u_{wc} \geq 0$  gives the joint LP

$$\begin{aligned}
\min \quad & t \\
\text{s.t.} \quad & t \geq \frac{1}{2} \sum_c (d_c^\top \lambda_{wc} + \mathbf{1}^\top u_{wc}) \quad \forall w, \\
& E_c^\top \lambda_{wc} + u_{wc} \geq \alpha_c (e_w - p) \quad \forall w, c, \\
& p \geq 0, \quad \mathbf{1}^\top p = 1, \quad u_{wc} \geq 0.
\end{aligned} \tag{S4.11}$$

For the three-policy known answer, A contains edges 1–3 and 2–3 with zero differences, while B contains edge 1–2 with  $x_1^B - x_2^B = 1/2$ . The compatible B vertices are

$$(1/2, 0, 0), (1/2, 0, 1), (1, 1/2, 0), (1, 1/2, 1).$$

The first two already give opposite unique target winners 1 and 3. Writing  $p = (p_1, p_2, p_3)$ , exact vertex evaluation yields

$$\mathcal{R}(p) = \frac{1}{8} \max\{1 - p_1, p_1 + 2p_2, 2 - 2p_1 - p_2, p_2\}. \tag{S4.12}$$

The maximum of the middle two terms is at least  $2/3 + p_2$ , with equality only when  $p_1 + p_2 = 2/3$ . This bound is uniquely minimized at  $p_2 = 0, p_1 = 2/3$ , giving  $p = (2/3, 0, 1/3)$  and regret  $1/12$ .

### S4.2 Minimum contextwise repair

Let the current route graph in context  $c$  have  $m_c$  connected components. One new pair type can merge at most two components, so at least  $m_c - 1$  additions are necessary for contextwise difference identification. If every cross-component policy pair is schedulable, any spanning tree on the component quotient graph supplies  $m_c - 1$  edges and is sufficient.

If only some pairs are allowable, form a quotient graph whose vertices are the current components and whose edges are allowable cross-component pairs. Repair is feasible exactly when this quotient is connected. With nonnegative pair costs, retain the cheapest policy pair for each quotient edge and choose a minimum spanning

tree. Consequently, when each new pair type belongs to only one context and every positive-target-weight context must be connected, the total minimum is

$$\sum_{c:\alpha_c>0} (m_c - 1), \quad (\text{S4.13})$$

provided every allowable quotient graph is connected.

In the three-policy example, context A is already connected and context B has components  $\{1, 2\}$  and  $\{3\}$ , so either B-context pair 1–3 or 2–3 closes the contextwise gap. A wholly unobserved B context has  $K$  singleton components and needs  $K - 1$  distinct B-context pair types for the same objective. Repeating an existing within-component pair does not change incidence rank. It may improve precision under a sampling model, but it cannot repair structural identification. Sufficiency also requires a named along-path stationarity/comparability condition: every edge in an identifying path must refer to the same stable context meaning. Connectivity alone does not supply that empirical bridge.

This count need not be minimal for identifying one winner at one boundary-constrained law, or when cross-context component offsets cancel. Connectivity and well-connected comparison design are prior art.

### S4.3 Simultaneous edge intervals

Finite samples replace the exact incidence equation by

$$\ell_c \leq E_c x^c \leq u_c, \quad 0 \leq x^c \leq 1. \quad (\text{S4.14})$$

Let  $\mathcal{X}(\ell, u)$  be the product of these contextwise feasible sets and let

$$\mathcal{M}(\ell, u) = \left\{ \sum_c \alpha_c x^c : (x^c)_c \in \mathcal{X}(\ell, u) \right\}. \quad (\text{S4.15})$$

The sharp bounds on target difference  $\mu_i - \mu_j$  are the minimum and maximum of that linear contrast over  $\mathcal{X}$ . Policy  $i$  is a possible winner if the same feasible system admits  $\mu_i \geq \mu_j$  for every  $j$ . It is certified as the unique winner only when the lower bound on  $\mu_i - \mu_j$  is strictly positive for every  $j \neq i$ .

If the intervals in (S4.14) jointly contain the population edge differences with probability at least  $1 - \alpha$ , then the population target belongs to  $\mathcal{M}(\ell, u)$  with at least the same probability. The projected difference bounds and any certified winner therefore inherit that coverage. This is a deterministic confidence-set projection: it adds no assumption and cannot repair invalid edge intervals.

The minimax action is also an LP. Define

$$A_c = \begin{bmatrix} E_c \\ -E_c \\ I \\ -I \end{bmatrix}, \quad b_c = \begin{bmatrix} u_c \\ -\ell_c \\ \mathbf{1} \\ \mathbf{0} \end{bmatrix}.$$

Dualizing the contextwise support functions gives

$$\begin{aligned} \min \quad & t \\ \text{s.t.} \quad & t \geq \frac{1}{2} \sum_c b_c^\top y_{wc} && \forall w, \\ & A_c^\top y_{wc} = \alpha_c (e_w - p) && \forall w, c, \\ & p \geq 0, \quad \mathbf{1}^\top p = 1, \quad y_{wc} \geq 0. \end{aligned} \quad (\text{S4.16})$$

Point intervals recover (S4.11). For the three-policy known answer, the program again returns  $p = (2/3, 0, 1/3)$  and regret  $1/12$ . A separate vertex-enumeration implementation agrees on that case and on connected, disconnected, and widened-interval controls.

One distribution-free input construction is available when each observed edge has  $n_e$  independent bounded half-credit outcomes  $Z \in [0, 1]$ . For  $M$  edges, Bonferroni–Hoeffding intervals use

$$\epsilon_e = \sqrt{\frac{\log(2M/\alpha)}{2n_e}}, \quad q_e \in [\hat{q}_e - \epsilon_e, \hat{q}_e + \epsilon_e] \cap [0, 1], \quad (\text{S4.17})$$

then map to  $d_e = 2q_e - 1$ . Across-edge independence is unnecessary for the Bonferroni step, but every edge interval must be valid for its own sampling unit. Clustered, sequential, or adaptively collected trials need a procedure valid for that design.

#### S4.4 Bradley–Terry connectivity corollary

Suppose exact target-context comparisons follow the Bradley–Terry law

$$q_{ij} = \sigma(s_i - s_j), \quad \sigma(z) = \frac{1}{1 + e^{-z}}. \quad (\text{S4.18})$$

For each observed common-context edge, the inverse link identifies  $s_i - s_j = \text{logit}(q_{ij})$ . If the comparison graph is connected, these differences identify all scores up to one common offset. The Borda ordering equals the score ordering: if  $s_i > s_j$ , then policy  $i$ ’s comparison probability exceeds policy  $j$ ’s against every third policy and also in their direct comparison. Equal scores give equal Borda values. Consequently the Borda-winner set is identified, and it is a singleton exactly when the identified largest score is unique.

If the graph has two or more components, each component retains its own additive offset. Shifting one component above another changes the global winner set without changing an observed edge. Thus, with unrestricted finite scores, the winner set is identified if and only if the graph is connected. A graph with  $m$  current components needs  $m - 1$  cross-component edges when such a repair is feasible;  $K - 1$  is the empty-graph case.

This is the standard connected-comparison-graph identifiability consequence for a latent difference model, not a new Bradley–Terry theorem. It also does not extend to strong stochastic transitivity alone, which is a broader model and does not make connected edge probabilities determine every score difference.

### S5. Same-box comparison with pairwise-lottery antecedents

Write the target margin matrix as  $M_{ij} = 2q_{ij} - 1$ , so every primitive edge lies independently in  $[-1/2, 1/2]$ . The four neighboring questions are not interchangeable.

| framework                     | uncertainty and comparator                                   | action/output on this example                     |
|-------------------------------|--------------------------------------------------------------|---------------------------------------------------|
| Observed maximal lottery      | Known routed matrix $M^0 = 0$ ; worst opposing lottery       | Every policy lottery is maximal                   |
| Incomplete equilibrium action | Equilibrium support in some or every completion              | Every action is possible; none is necessary       |
| Robust maximal lottery        | $\max_p \min_M \min_q p^\top M q$                            | Unique uniform $p$ ; worst margin $-(K - 1)/(2K)$ |
| P2 Borda regret               | Regret to the Borda-best policy in the same compatible world | Unique uniform $p$ ; regret $(K - 1)/(4K)$        |

For fixed  $p$ , a pure opponent  $j$  and the completion  $M_{ij} = -1/2$  for every  $i \neq j$  give

$$\min_M \min_q p^\top M q = -\frac{1}{2} \max_j (1 - p_j) = -\frac{1}{2} (1 - \min_j p_j). \quad (\text{S5.1})$$

The robust maximal-lottery value is therefore maximized uniquely by the uniform lottery. Its margin value is  $-(K-1)/(2K)$ , equivalently worst win probability  $(K+1)/(4K)$ . The selected lottery happens to agree with P2 on this symmetric box, but the loss, comparator, value, and interpretation differ.

The zero matrix is one completion, so every action is possible in an equilibrium support. For any queried action, another completion can make a different action the unique Condorcet winner and unique maximin action; no action is necessary. This comparison follows Brandl, Brandt, and Seedig; Brill, Freeman, and Conitzer; and Khalaf et al. It is not a reproduction of those papers or evidence of novelty.

## S6. Prior-art boundary

The deterministic value in S2 is the marginal interval width

$$U - L = \frac{K-1}{2K}, \quad L = \frac{K+1}{4K}, \quad U = \frac{3K-1}{4K},$$

and is already recovered by Stoye’s arbitrary-finite-treatment interval model. General robust randomization, zero-sum reduction, optimizer-set analysis, and nonunique minimax decisions are also established work. The bounded surviving objects are the edge-coupled randomized value and uniqueness in S2 and the weighted exact face in S3, each for this constructed compatible class. Absence of a formula from the bounded search is not evidence of novelty.

## S7. Public-evidence and diagnostic-reference details

### S7.1 Evidence layers

The evidence labels are cumulative only within a specified unit: *source-described* means a fixed primary source states the feature; *artifact-reconstructable* means pinned code or data support it; *artifact-partial* means only a stated subset is reconstructable; and *realized-law verified* requires version-bound assignment and lifecycle records. “Absent” always means absent from the fixed public sources, not privately absent.

H251 fixes the three AnkIle R5 repositories at revisions 15d4190..., cd5d0d..., and d2d109...; RoboArena DataDump\_07-17-2026 at 7931db8...; and TRI Dryad version 4. The AnkIle rectangles contain all 450 declared policy-state cells. RoboArena contains 104 of 210 possible label pairs: the 20 pair-eligible labels are connected, while one unpaired label is isolated. TRI’s 20 CSVs omit the published test-bundle join; three cells need a narrow trailing-apostrophe parse repair, and two released rates disagree with their arrays and counts. Separate Python and Ruby implementations reproduce these facts.

The AnkIle released-panel margins are three, one, and three successes. In a worst-case retained-round sensitivity, replacing a matched round by an arbitrary three-policy binary outcome vector can reverse the routing, marker, and square winners in two, one, and two rounds, respectively. Exact subset enumeration verifies those minima. This bounds fragility; it does not show that any round was replaced. The released configuration permits incomplete-round reruns for routing and marker and disables them for square.

| system    | favorable source evidence                                                             | reconstructed public layer                                                                                                              | unresolved layer                                                                        |
|-----------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Ankle R5  | Three policies on the same 50 declared states per task; randomized within-round order | Complete rectangles, exact outcomes, fixed artifacts, state manifests, submitted-round lists, and one/two-round replacement sensitivity | Whether any retained round replaced an earlier attempt and why; population sampling law |
| RoboArena | Random pair sampling; policy-identity blinding                                        | Pinned 3,883 sessions; 21 labels; 104 pair edges; complete public schema                                                                | Deployed weights, pool epochs, context bridge, exact resets, robot and attempt lineage  |
| TRI LBM   | Blind randomized matched test bundles in the paper                                    | Version-4 outcome arrays and margins; two rate inconsistencies                                                                          | Bundle/trial/order/reset, session/robot, retry, and immutable policy-version join       |

Five earlier systems supply useful design or metadata evidence but do not add an artifact-reconstructable common-context comparison.

| system      | favorable source evidence                                                | reconstructed public layer                                        | unresolved layer                                                            |
|-------------|--------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------|
| UMI-Bench   | Common finite frame; reset and trial records described                   | Demonstration corpus is not the evaluation runner or trial ledger | Execution ledger and realized law                                           |
| RoboDojo    | Common layouts and roster; blinded scoring                               | Frame, roster, execution path, trial identity; reset partial      | Historical realization and accepted-state evidence                          |
| ArmnetBench | All policies per cell; independent placement; checkpoint release claimed | Public trials and 84 resolving repositories                       | Exact revisions, execution order, reset law, realized state, physical block |
| PhAIL       | Context first; <b>BalancedSampler</b> ; operator blinding                | Sampler code, context fields, labels, 594 episode identities      | Realized draws, historical revision, reset lineage, operator/physical block |
| RRC 2020    | Complete retained runs; run type owner-confirmed from ID format          | Job/start/robot index and structural audit                        | Excluded attempts, policy/team join, retry parent, attempt-level selection  |

## S7.2 Diagnostic metadata sensitivities

The values below are Monte Carlo reference-tail fractions under explicitly chosen diagnostic laws. They are not p-values calibrated by an identified physical assignment, sampling, session, or dependence law.

| analysis                       | diagnostic reference                                   | draws  | decision boundary                               | exposure            |
|--------------------------------|--------------------------------------------------------|--------|-------------------------------------------------|---------------------|
| State simulation               | Independent-uniform state simulation; fixed statistics | 49,999 | Joint tail/effect gates                         | Fixed before result |
| Clock-regime state permutation | State permutation within clock regime                  | 49,999 | Pooled 0.01/10% gate; two 0.005 secondary gates | Result-exposed      |
| Clock-regime label permutation | Label permutation within clock regime                  | 49,999 | Pooled 0.01/0.10 gate; two Bonferroni tests     | Result-exposed      |
| UTC-date label permutation     | Label permutation within UTC date                      | 49,999 | Same pooled/secondary gates                     | Result-exposed      |

The state simulation and clock-regime state permutation miss their fixed material gates. The clock-regime label permutation crosses one secondary gate but misses the pooled effect gate. Under the UTC-date reference, the pooled and regime-1 excesses contract to 0.01205 and 0.03704, with tail fractions 0.52872 and 0.23688. These diagnostics motivate a metadata request; they do not establish independence, assignment, session, carryover, outcome uncertainty, or performance.